

Team D: Catalyst

Mentors: George Kantor &
Newell Washburn (Cloud Lab)

Meet the Team



Srishti Gupta



Shua Halle



Andrew Noble



Risha Ranjan



Aman Tambi

Table of contents

- (1) Needs & Use Case
- (2) Requirements
- (3) System Architectures
- (4) Trade Studies
- (5) Project Management
- (6) Risks

1

Needs & Use Case

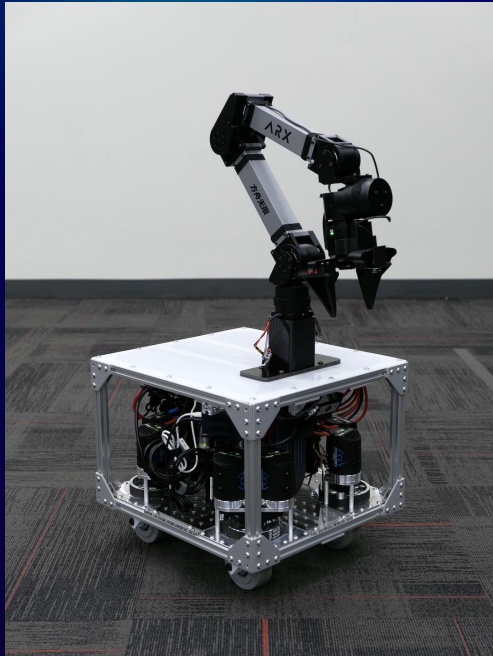
Needs

- Machines operate in isolation
- Humans must transfer materials
- Causes delays and error risk
- No true 24/7 operation
- Manual steps create bottlenecks



Our Proposed Solution

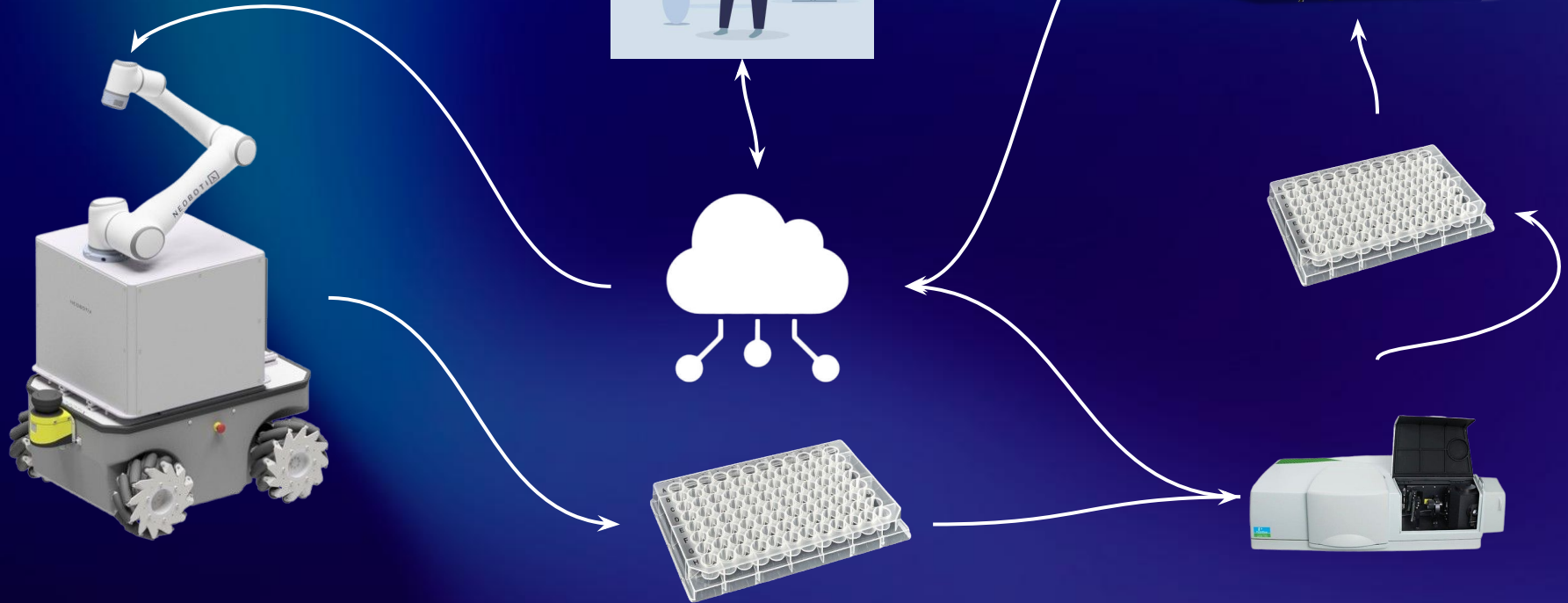
A mobile manipulator to autonomously move lab materials between automated machines.



Advantages of automating this process:

1. Allows remote experimentation
2. Reduces human error and risk of contamination
3. 24/7 operation which generates significantly more experimental data:
 - a. Ideal for training AI models
 - b. Enables faster drug discovery cycles in drug development
 - c. Enables rapid scaling of experimental cycles

Use Case



2

Requirements

Requirements

Functional Requirements

Localize wellplate

Grasp wellplate

Pick & place wellplate in Instrument

Avoid obstacles

Localize within lab space

Navigate between instruments

Nonfunctional Requirements

Safe operation with lab personnel

Does not spill well plate liquid

Fits between lab aisles

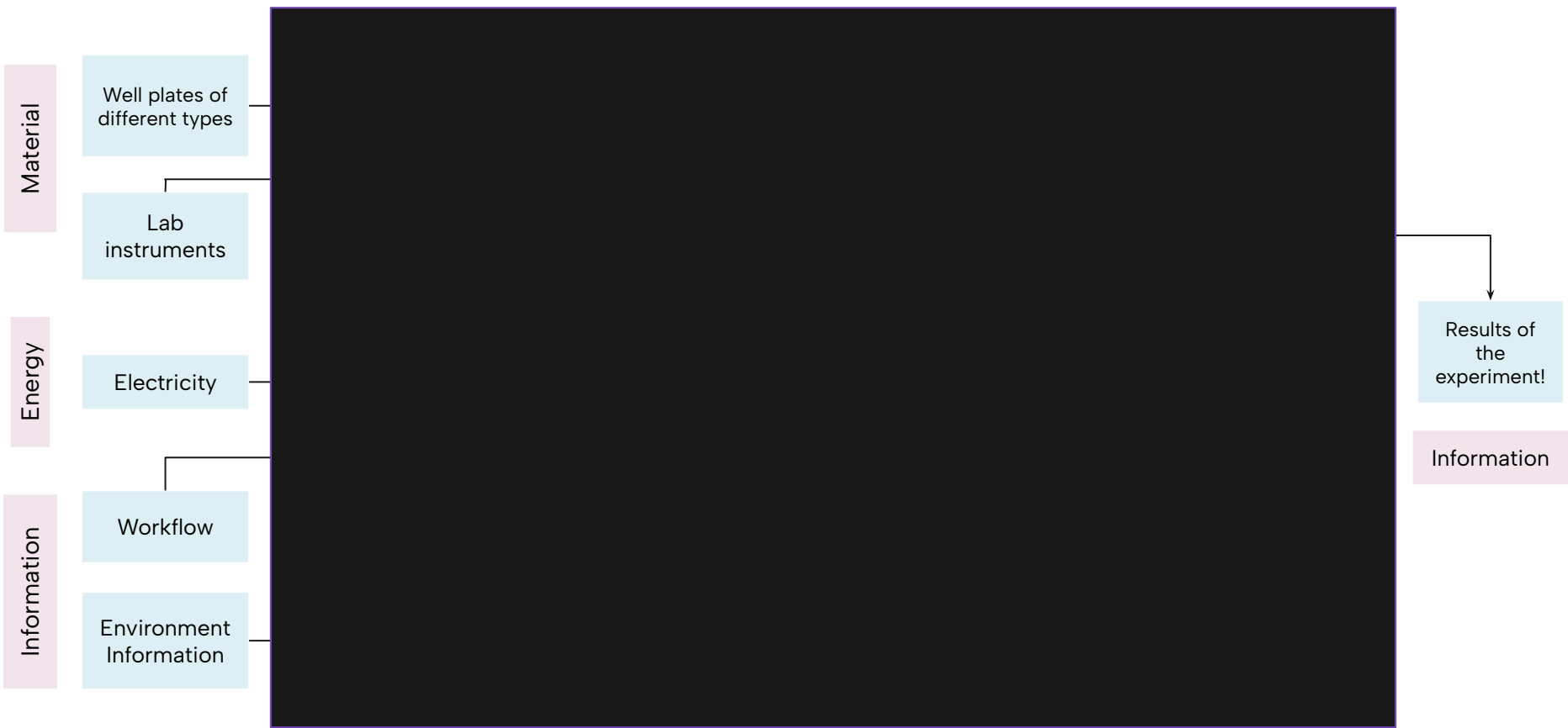
Can be easily recharged

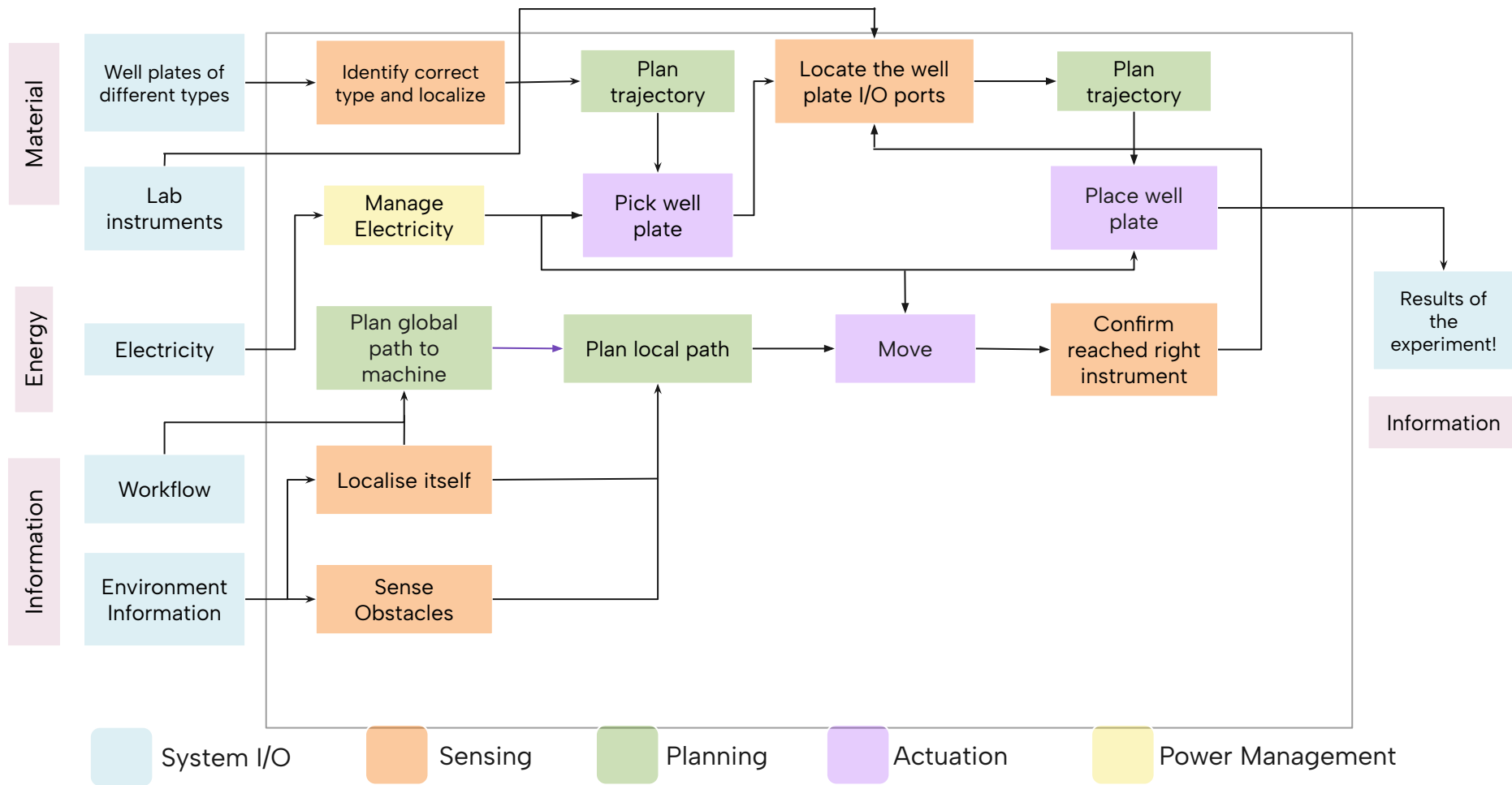
Performance Requirements

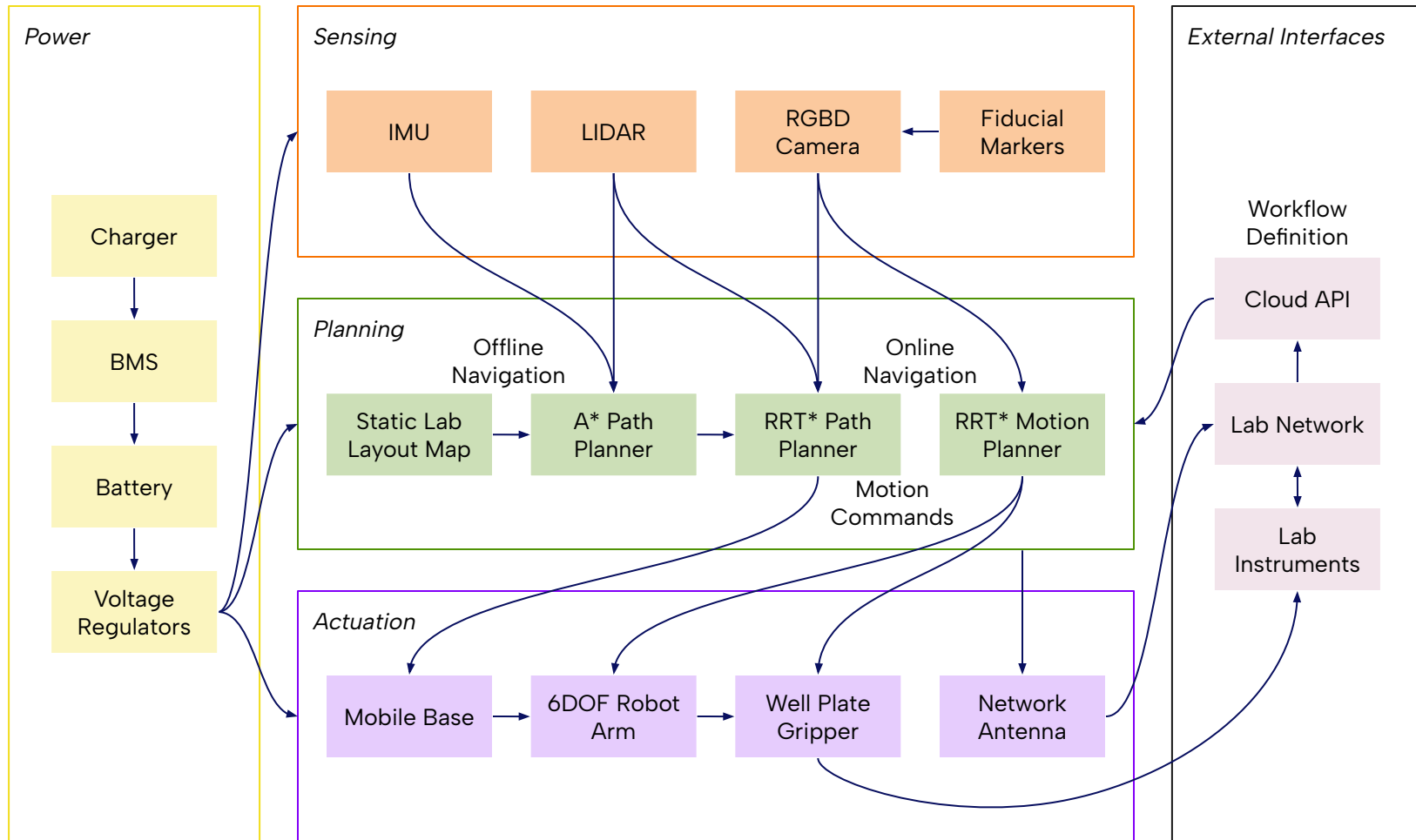
Requirement	Metric
Pick & place well plates in instrument	90% reliability
Localize with respect to instrument	± 10 cm of nominal location
Transit speed	20 cm/s
Pick/place duration	1 min

3

Functional and Cyberphysical Architecture





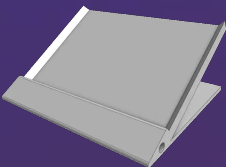
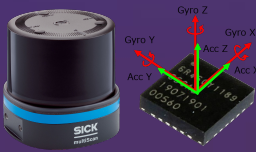
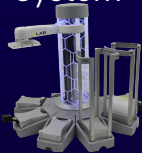
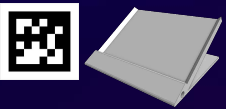
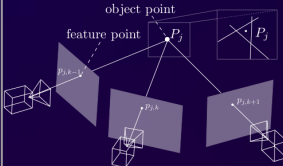


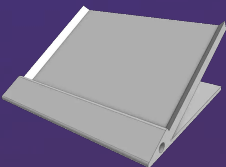
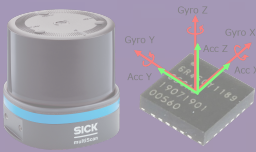
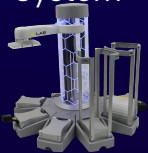
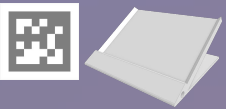
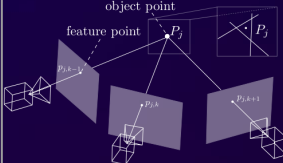
4

Trade Studies

System Level

Spec/Criteria	Weight	Human-Based	Fixed Workcell	Mobile Manipulator	Gantry Automation
Works Autonomously	25%	1	3	5	4
Flexible Across Instruments	20%	5	2	5	2
Scalability & Repeatability	25%	2	4	4	5
Cost	10%	3	3	3	2
Speed of Operation	20%	3	4	4	5
Total Score	100%	2.65	3.25	4.35	3.85

Grasp wellplate	Localize wellplate	Detect obstacles	Localize within lab space	Drive Mechanism
Adaptive Multi-Finger 	RGB + Fiducial Marker 	Single-Beam Time-of-Flight 	Magnetic Tape + Hall Sensor 	Skid Steer 
Vacuum 	RGBD Camera + CV 	360° LIDAR 	Fiducial Marker 	Diff Drive 
Parallel-Jaw 	Locating Fixture 	RGBD Camera 	LIDAR + IMU 	Holonomic Omni-wheel 
Tool-Changing System 	Markers with Fixture 	RGB Camera + CV YOLO 	Visual Odometry 	Holonomic Swerve Drive 

Grasp wellplate	Localize wellplate	Detect obstacles	Localize within lab space	Drive Mechanism
Adaptive Multi-Finger 	RGB + Fiducial Marker 	Single-Beam Time-of-Flight 	Magnetic Tape + Hall Sensor 	Skid Steer 
Vacuum 	RGBD Camera + CV 	2D LIDAR 	Fiducial Marker 	Diff Drive 
Parallel-Jaw 	Locating Fixture 	RGBD Camera 	LIDAR + IMU 	Holonomic Omni-wheel 
Tool-Changing System 	Markers with Fixture 	RGB Camera + CV YOLO 	Visual Odometry 	Holonomic Swerve Drive 

Localization Method (Lab)

Spec/Criteria	Weight	Magnetic Tape	Fiducial Marker	LIDAR + IMU	Visual Odometry
Localisation Accuracy	20%	5	5	4	3
Flexibility to map changes	20%	1	3	4	4
Maintenance effort	15%	1	3	4	3
Sensor Cost	10%	5	4	1	3
Scalability	20%	1	3	5	4
Works in Obstructed lab spaces	15%	2	3	5	3
Total Score	100%	2.35	3.5	4.05	3.4

Gripper Strategy

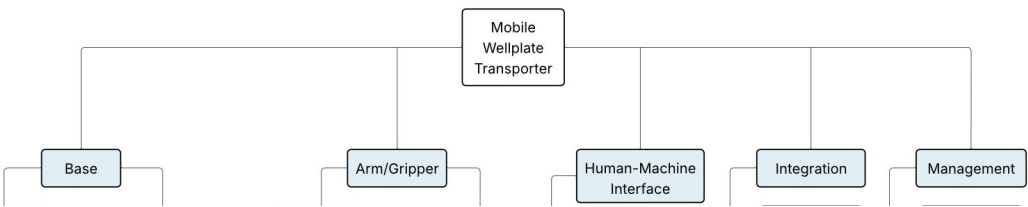
Spec/Criteria	Weight	Adaptive Multi-Finger	Vacuum	Parallel-Jaw	Tool-Changing System
Compatibility with Labware	25%	3	2	4	5
Customizable Fingers	20%	1	1	5	5
Ease of Integration	20%	3	2	5	1
Cost	10%	2	5	4	3
Positioning Precision	25%	3	2	4	5
Total Score	100%	2.5	2.1	4.4	4

Perception: Docking & Plate Detection

Spec/Criteria	Weight	2D RGB + ArUco	RGB-D Camera	Mechanical Fixture only	Hybrid Fixture + Camera
Precision of Alignment					
Robustness to Lighting					
Complexity of Integration					
Scalability to new devices					
Pose Estimation					
Well Plate Detection					
Total Score					

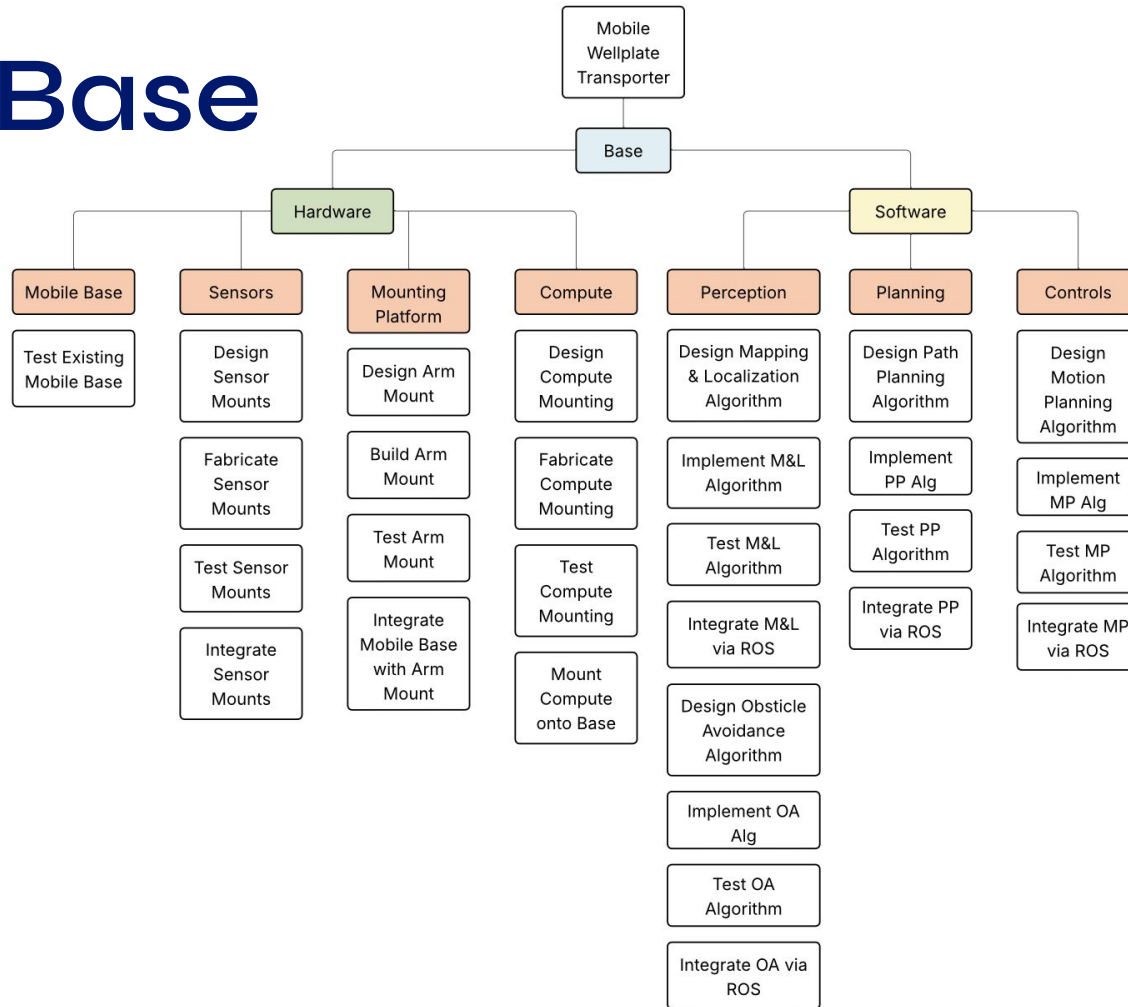
5

Project Management

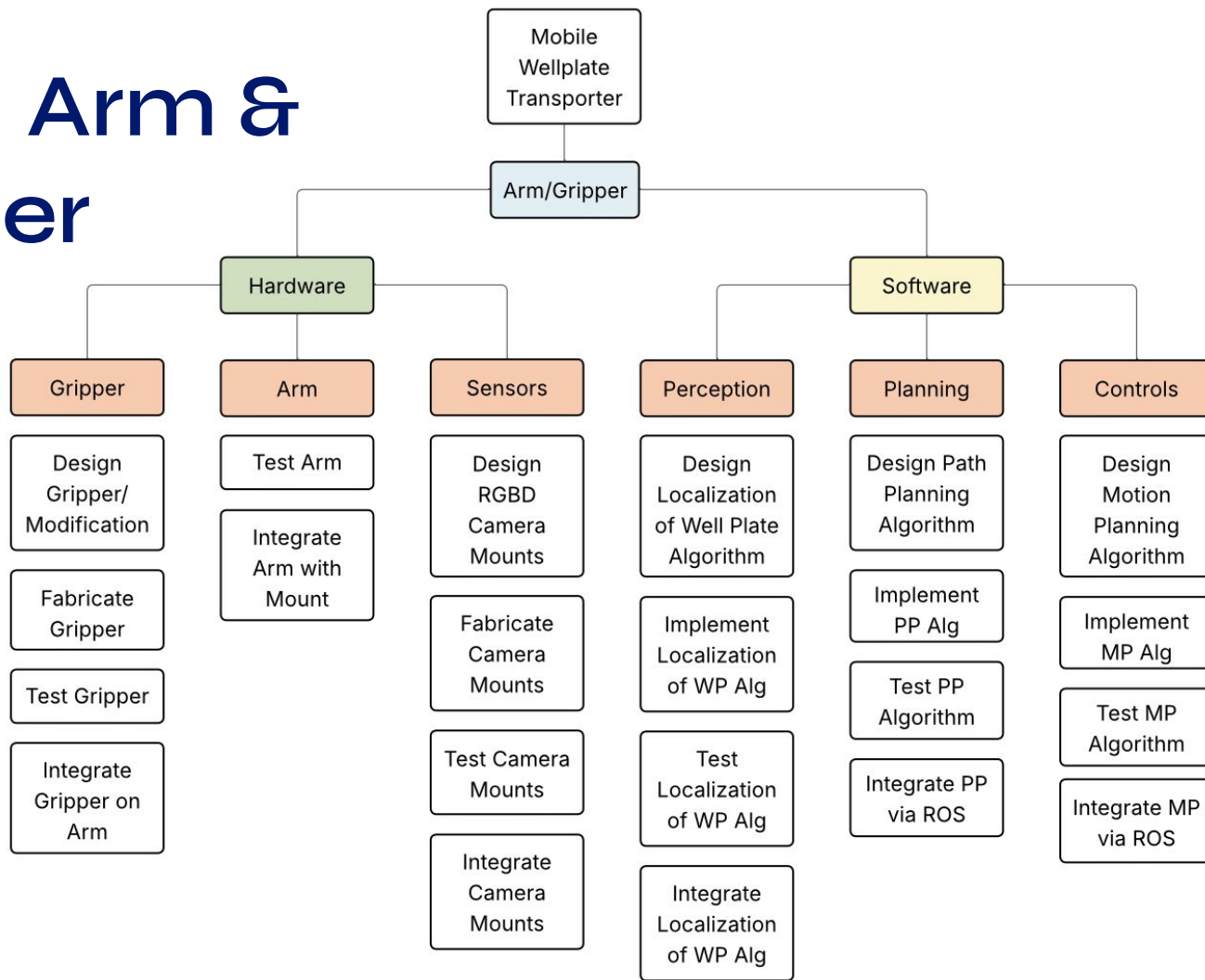


Work Breakdown Structure

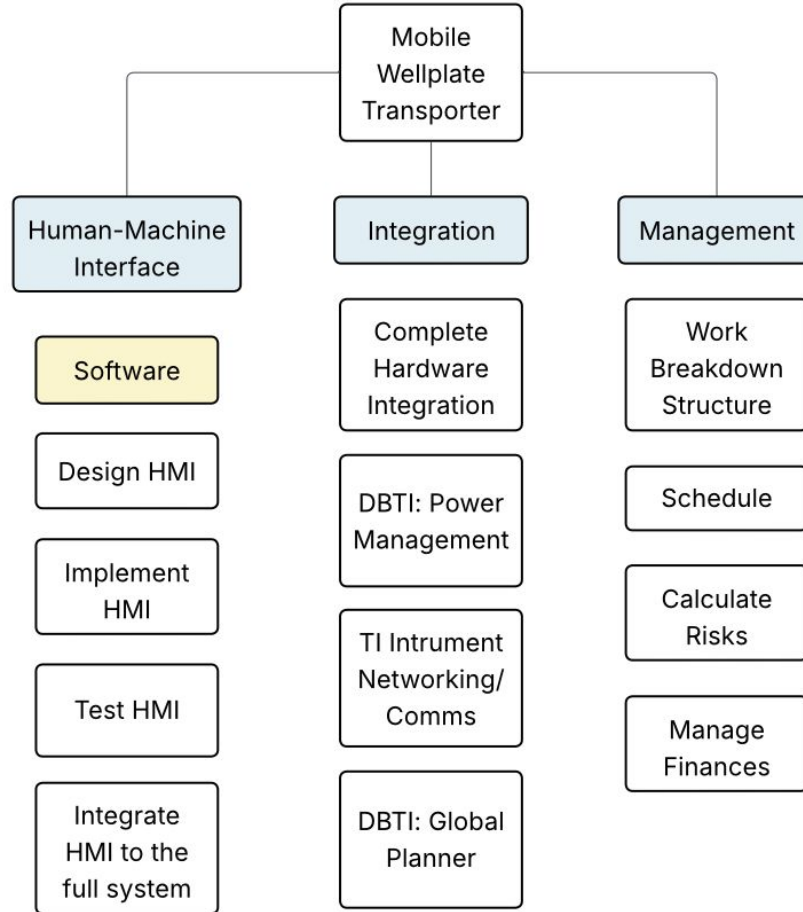
WBS: Base



WBS: Arm & Gripper



WBS: HMI, Integration, Management



Planned SVD

Base

Base starts a specified distance from a one type of instrument and successfully navigates to the instrument with ± 10 cm accuracy.

Manipulator

Stationary manipulator successfully picks and places wellplates from a single type of lab instrument

Fall validation day (tentative)

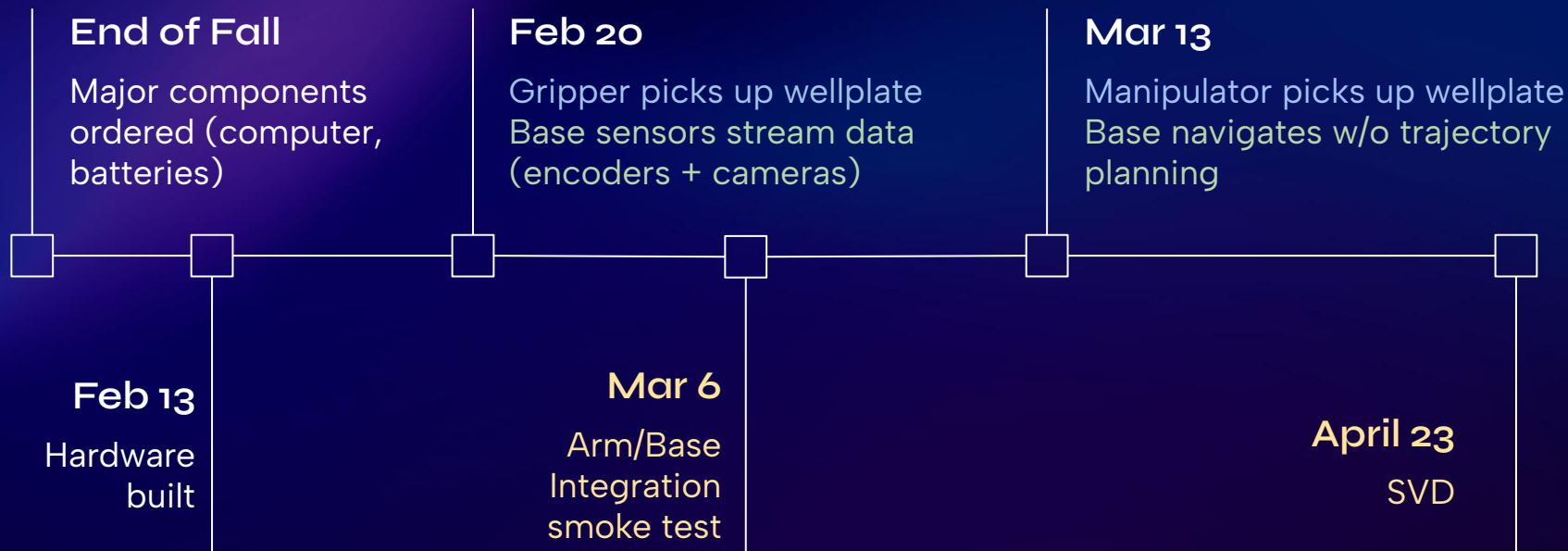
- integrated system
- networked with instruments
- Multi-instrument

Major Milestones

Arm Development

Base Development

Milestone



Q1 2026

Q2 2026

January

February

March

April

May

▶ Base Sensor Mounts - HW 3 ㉟

▶ Arm - Base Integration HW 4 ㉟

◇ Hardware Fabricated
Due Feb 13, 2026

▶ Arm - HW 2 ㉟

▶ Arm End Effector - HW 4 ㉟

▶ Base Compute Mount - HW 4 ㉟

▶ Base Controls - SW 3 ㉟

▶ Base Path Planning - SW 3 ㉟

▶ Base SLAM - SW 3 ㉟

▶ Base Obstacle Avoidance - SW 3 ㉟

▶ Arm Controls Algorithm - SW 3 ㉟

▶ Arm Trajectory Planning - SW 3 ㉟

▶ Arm Wellplate Localization -
SW 2 ㉟◇ Arm - Base Integration Sm...
Due Mar 6, 2026◇ PDR Presentation
Due Mar 10, 2026◇ Spring Validation Day
Due Apr 23, 2026◇ CDR Presentation
Due Apr 27, 2026

6

Risks

Risks

Risk	Likelihood (/5)	Consequence (/5)	Mitigation	Worst-case
Difficult cloud lab access	2	3	Communicate with sponsor	Use mock setup
Manipulation reliability harder than expected	2	5	Early gripper testing	Ask for SME advice
Integration is harder than expected	3	4	Arm-Base Smoke Test2	Change scope to stationary arm
Cannot network with instruments	2	2	Communicate with Cloudlab/NREC	Change scope to remove networking
We damage hardware	3	3	Work in pairs to prevent mistakes	Change scope

Risks

1. Cloud lab is unavailable as testing space (L2, C2)
 - Mitigation: written confirmation
 - Issue: mock setup in george's lab
2. No mobile platform available (L2, C4)
 - Mitigation: use husky
 - Issue: eliminate mobile platform subsystem
3. LIDAR insufficient for localization (L4, C2)
 - Mitigation: perform thorough tests to verify accuracy
 - Issue: use high resolution cameras
4. Manipulator not accurate enough (L1, C5)
 - Mitigation: Early test
 - Issue: get external support
5. Subsystem integration delays (L4, C4)
 - Mitigation: Overestimate time needed for integration, integrate in parallel with development
 - Issue: Remove the mobility subsystem
6. Machine networking integration issues (L2, C2)
 - Mitigation: contact NREC for documentation
 - Issue: change scope by eliminating instrument interaction
7. Damage hardware or instruments during development (L3, C3)
 - Mitigation: Work in small groups to self-supervise, test with mock hardware to reduce danger
 - Issue: Keep buffer in budget for emergencies
8. Low budget

Thanks!

Any questions?